Prob.1. (2 & T, Ch. 7, 8.7)

Suppose that f is defined & cont-s on $[-\pi, \pi]$. State the conditions under which the F.S of f converges to $f(x)$ for <u>every</u> $x \in [-\pi, \pi]$.

Sol-n:

Recall the following properties of Fourier series:

(i) F-S is periodic on $[-\pi, \pi]$.

(ii) What f-ns have the F-S? Answer: ALL integrals that are finite:

$$|a_n \cos(n\theta) + b_n \sin(n\theta)| \leq |a_n| + |b_n|$$

(iii) When does the F-S converge to its f-n $f(x)$?

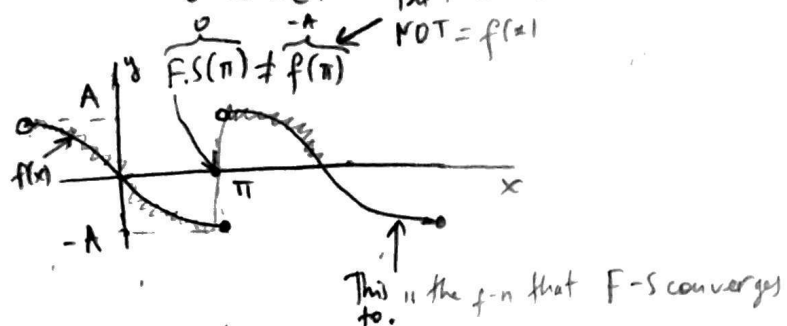
If $f(x)$ is <u>piecewise smooth</u> on $-\pi \leq x \leq \pi$;

<u>Piecewise smooth</u>: The interval of interest can be broken up to pieces where $f(x) \in C^1(I_i)$ where $\cup I_i = \Omega$.

(iv) <u>Matching end points</u>, that is

$$f(-\pi) = f(\pi)$$

e.g.



(iii) answers the question, (iv) can be extra supporting. ∎

Prob.2. (2 & T, Ch. 7, 8.10)

Derive the F-S for the ↓ f-ns & then sketch the graph of the limit to which the series converges for all x.

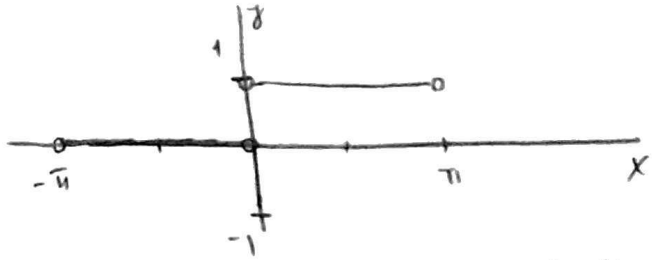
(i)

$$f(x) = \begin{cases} 0, & \text{for } -\pi < x < 0 \\ 1, & \text{for } 0 < x < \pi \end{cases}$$

Sol-n [Preamble].

(1) $f(x) = \begin{cases} 0, & \text{for } -\pi < x < 0 \\ 1, & \text{for } 0 < x < \pi \end{cases}$

(2)

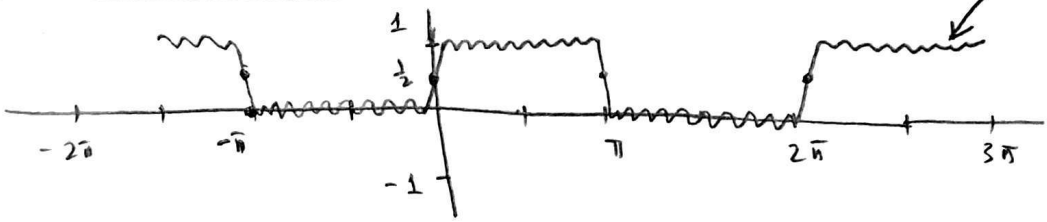


Thus by (3), we have

if $n = 2k$ $\Sigma \cdot = 0$. So let $n = 2k-1$, $k = 1, 2.$

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \left[0 \cdot \cos(nx) + \left(\frac{1 - \cos(\pi n)}{\pi n} \right) \cdot \sin(nx) \right] = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{1 - \cos(\pi n)}{\pi n} \cdot \sin(nx)$$

(10) $$\boxed{f(x) = \frac{1}{2} + \sum_{k=1}^{\infty} \frac{2}{(2k-1)\pi} \cdot \sin[(2k-1)x]}$$



f-n that F-S converges to $\neq f(x)$.

(c)